

Project Completion Report

Hydra Point-of-Sale: Accept ADA, CNTs in Retail & Invoicing

Field	Value
Project Name	Hydra Point-of-Sale: Accept ADA, CNTs in Retail & Invoicing
Project Number	1400065
Challenge	F14: Cardano Open: Developers
Project Manager	Rahul Konudula
Project Start Date	November 24, 2025
Project Completion Date	June 1, 2026
Repository	github.com/Trivolve-Tech/Hydra-Point-of-Sale-Accept-ADA-CNTs-in-Retail-Invoicing
Documentation site	trivolve-tech.github.io/Hydra-Point-of-Sale-...

1. Deliverables

The Hydra Point-of-Sale project delivered a **self-hosted, open-source point-of-sale + invoicing toolkit** that accepts **ADA and Cardano Native Tokens (CNTs)** with **Hydra Layer-2 settlement on Cardano mainnet**. The system ships a single Next.js application that serves both a merchant dashboard and a customer payment page, plus a per-customer Hydra-head orchestrator and a sanitised mainnet docker-compose stack. Real ADA, real Hydra heads, real CIP-30 wallets — and rolled out for live retail payments at the **Sova Bar (Da Nang) launch event**.

Single URL hosting all outputs

Documentation site (GitHub Pages): <https://trivolve-tech.github.io/Hydra-Point-of-Sale-Accept-ADA-CNTs-in-Retail-Invoicing/>

Off-chain evidence — code, documentation, and releases

Output	Link
Source repository	github.com/Trivolve-Tech/Hydra-Point-of-Sale-Accept-ADA-CNTs-in-Retail-Invoicing
v1.0.0 release (mainnet baseline)	Releases — v1.0.0
v2.0.0 release (hybrid-custody mainnet rewrite — better retail + business UX)	Releases — v2.0.0
Project README	README.md
Changelog (SemVer)	CHANGELOG.md
Launch-event Feedback	docs/FEEDBACK.md
Hybrid-custody architecture	docs/architecture/hybrid-custody.md
Mainnet deployment guide	docs/ops/mainnet-deployment.md
Key custody policy	docs/ops/key-custody.md
Incident playbook	docs/ops/incident-playbook.md
<code>--deposit-period</code> tuning	docs/ops/deposit-period-tuning.md
Merchant + customer + API surface	merchant-pos/src/server/
In-app demo videos	docs/demo-videos/

Specification PDFs (in-repo)

PDF	Direct link
Requirements Specification Document	Hydra_Point-of-Sale_(PoS)__Invoicing_-_Requirements_Specification_Document.pdf
Technical Architecture & Design	Hydra_Point-of-Sale_(PoS)__Invoicing_Technical_architecture_and_design.pdf

Component inventory (v2.0.0)

Path	Purpose
<code>merchant-pos/</code>	Next.js 15 app serving both the merchant dashboard (/) and the customer payment page (/customer); HTTP API under /api/*
<code>merchant-pos/src/server/</code>	<code>hydra-node</code> WS session manager, per-customer L2 spend-key profile module, Cardano tx builders, payment router, in-head signer, L1 submitter, Postgres data layer (Drizzle ORM)
<code>docker/docker-compose.mainnet.yml</code>	One-shot bring-up of Postgres + <code>cardano-node</code> + the Next.js app on mainnet
<code>docker/docker-compose.head.tpl.yml</code>	Per- (merchant, customer) Hydra-head template the orchestrator instantiates
<code>hydra-config/protocol-parameters.json</code>	In-head ledger protocol parameters used when building L2 transactions
<code>scripts/derive-merchant-addresses.ts</code>	Derive operator BIP-39 wallet's per-head fee-funding addresses
<code>scripts/reconcile.ts</code>	Cross-check DB intent state vs <code>hydra-node</code> /snapshot/utxo
<code>.github/workflows/release.yml</code>	Tag-triggered GitHub Release workflow

On-chain context

The project ships **mainnet-ready code** that runs against **Cardano mainnet** via Blockfrost using `hydra-node` 1.3.0 and CIP-30 wallets (Vespr, Eternl, Lace, Nami, Flint, ...). Because Hydra heads are **operator-instantiated** per (merchant, customer) pair, there is no single canonical contract address to publish — each deployment opens its own heads and the operator's wallet addresses are what hit chain. The architecture and the L1/L2 split are documented in [docs/architecture/hybrid-custody.md](#) : **L1 commits are non-custodial** (wallet-signed by the customer's CIP-30 wallet), **L2 spends are server-signed** with a per-customer ed25519 key, AES-256-GCM encrypted at rest.

Open source status

Yes — MIT licensed. Single [LICENSE](#) at the repo root.

Testing performed

- **Testnet pilot** — Hydra Head L2 payment routing with automatic L1 fallback across Cardano **Preprod + Preview** (v0.2.x line, see [CHANGELOG.md](#)).
- **Mainnet bring-up** — verified end-to-end via `docker compose --env-file docker/.env.mainnet -f docker/docker-compose.mainnet.yml up -d` , including the orchestrator-driven head-open path (`POST /api/heads/open`) and the customer commit path (`POST /api/heads/[id]/commit-deposit` → `POST /api/intents/[tx]/pay`).
- **Reconciliation tooling** — `scripts/reconcile.ts` cross-checks DB intent state against `hydra-node /snapshot/utxo` .
- **Determinism fix verified on mainnet** — `csl.FixedTransaction.sign_and_add_vkey_signature` preserves the original body bytes (re-serialising `TransactionBody` produced subtly different bytes and broke the body hash); this was caught and fixed before mainnet rollout (see [CHANGELOG.md](#) — Changed).
- **Live retail validation** — the Sova Bar (Da Nang) launch event drove real-customer retail payments end-to-end through the v2.0 stack.

User feedback

The launch-event feedback artefact at [docs/FEEDBACK.md](#) captures **four attendee testimonials** from the Sova Bar (Da Nang) launch — one from the venue partner (Apparao) and three from retail customers (Bella, Charan, Maria). Highlight: *"We love it — it is processing payments in sub-seconds. This is the best crypto PoS system I've used so far."* — **Apparao, Partner, Sova Bar**.

Visual evidence

- **In-app product demo videos** — [docs/demo-videos/](#): merchant full checkout, customer payment, Hydra L2 merchant + customer flows.
- **Community launch-event photos and videos** — [Google Drive folder](#) (Sova Bar, Da Nang).

2. Usage

Hydra Point-of-Sale is **live at Sova Bar (Da Nang)** for real retail payments and is **self-hostable** by any operator who wants to accept ADA and CNTs with Hydra Layer-2 settlement on Cardano mainnet.

Who uses it and how they interact

- **Retail merchants** run `docker compose --env-file docker/.env.mainnet -f docker/docker-compose.mainnet.yml up -d` to bring up the full stack (Postgres + cardano-node + Next.js app). The merchant dashboard at `/` opens heads, creates payment intents, and watches settlement state; the customer page at `/customer` is where the payer connects their CIP-30 wallet and pays.
- **Retail customers** open the customer page on their phone, connect a Cardano wallet (Vespr, Eternl, Lace, Nami, Flint...), and pay in **ADA or CNTs**. The customer sees a **six-step commit progress**

stepper with live elapsed/ETA timers and per-step status, then sub-second perceived settlement on the in-head L2 path.

- **Forks / self-hosters** clone the repo, follow [docs/ops/mainnet-deployment.md](#), and run on their own infrastructure under MIT.

Key actions completed

- **v1.0.0 mainnet release** published ([2026-05-24](#)).
- **v2.0.0 hybrid-custody mainnet rewrite** published ([2026-06-01](#)) — shipped with **a better retail-customer experience** (smoother CIP-30 flow, six-step progress stepper, sub-second perceived in-head L2 settlement) and **a better business experience** (single Next.js app, per-customer Hydra-head orchestration, sanitised mainnet env template, ops playbooks).
- **Sova Bar (Da Nang) live launch** — real customers paid in ADA / CNTs on Cardano mainnet, with attendee testimonials captured in [docs/FEEDBACK.md](#).
- **Public docs site live** — GitHub Pages serving deployment, integration, and API reference content.

Evidence of engagement

- Releases page (downloadable, installable by the public): [Releases](#).
 - Docs site: [GH Pages](#).
 - Launch event media: [Google Drive folder](#).
 - Attendee testimonials: [docs/FEEDBACK.md](#).
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3. Impact

Measurable value created (before → after)

Dimension	Before	After this project	Source
Open-source ADA + CNT PoS using Hydra L2 on Cardano mainnet	Did not exist as a self-hostable product	MIT-licensed, mainnet-deployed stack with hybrid-custody design	Repo + v2.0.0 release
Retail-customer settlement latency	Card networks: seconds-to-minutes round-trip	Sub-second perceived in-head L2 settlement (cite Apparao's testimonial)	docs/FEEDBACK.md
Self-custody for retail payers	Crypto cards intermediate the payment	Direct CIP-30 wallet → merchant (Charan: <i>"nice to use my crypto directly without cards"</i>)	FEEDBACK.md
Operator deployment cost	DIY assembly of cardano-node + hydra-node + payments stack	One-shot docker-compose with sanitised mainnet env template + ops playbooks	docker/docker-compose.mainnet.yml , docs/ops
CIP-30 wallet compatibility for Hydra L2	Limited / not productized	Hybrid-custody architecture works with current CIP-30 wallets (which refuse to sign for L1-absent UTxOs)	docs/architecture/hybrid-custody.md

Performance & reliability

- **Sub-second perceived L2 settlement** at the live launch event (verbatim merchant partner feedback).
- **L1 commit/close cost paid once per head**, not per payment — economical for repeat retail customers.
- **--deposit-period raised from 60 s to 300 s** (verified failing at 60 s on mainnet) so the snapshot leader has time to incorporate the deposit before it expires (`CHANGELOG.md` — Changed).
- **Body-hash determinism** fixed via `cs1.FixedTransaction.sign_and_add_vkey_signature` so L2 signing always matches the original body bytes.
- **Reconcile script** (`scripts/reconcile.ts`) catches drift between DB intent state and `hydra-node` / `snapshot/utxo` .
- **HPOS_KEY_SECRET required** — the customer-profile module refuses to start without it; **no defaults** for `WALLET_SEED_PHRASE` , `BLOCKFROST_KEY` , `SITE_PASSWORD_MERCHANT` , or `HPOS_KEY_SECRET` (fail-closed).

Cardano ecosystem benefit

- **First open-source ADA + CNT point-of-sale stack** built on Hydra L2 for **Cardano mainnet** — gives any Cardano-aligned venue a turnkey path to accept crypto natively.
- **Reusable per-head docker orchestrator** + sanitised env templates lower the operational barrier for other Cardano builders integrating Hydra heads.
- **CIP-30 wallet compatibility** advances mobile / browser-wallet Hydra adoption (the hybrid-custody design is the only model that works today against wallets that won't sign for L1-absent UTxOs).
- **Operational maturity** — published playbooks for [deployment](#), [key custody](#), [incidents](#), and [--deposit-period tuning](#).

Recognition

- **Sova Bar (Da Nang)** partnered to deploy the system for live retail payments.
- Four launch-event attendee testimonials captured in [docs/FEEDBACK.md](#) .

4. Sustainability

This project is **ongoing** as a maintained, public, MIT-licensed open-source codebase. The v2.0.0 hybrid-custody mainnet release is the milestone cut; the repository remains the canonical home and continues to evolve.

Maintenance model

- **Repository:** github.com/Trivolve-Tech/Hydra-Point-of-Sale-Accept-ADA-CNTs-in-Retail-Invoicing — issues / PRs accepted.
- **Versioning:** SemVer with a maintained [CHANGELOG.md](#) (Added / Changed / Removed / Security sections per release).
- **Release automation:** `.github/workflows/release.yml` builds GitHub Releases on `v*` tags.
- **Docs site:** GitHub Pages, auto-deployed (rebuilds when `docs/` changes).
- **Operational guides:** deployment, key-custody, incident-playbook, deposit-period-tuning under [docs/ops/](#) keep operator knowledge in-tree.

Revenue model

The project is free and MIT-licensed — operators bear their own hosting + Blockfrost costs and pay their own Cardano L1 fees for head open/close. There is no protocol fee, no token, no central treasury attached. Ongoing maintenance, additional CNT / wallet integrations, and merchant-facing features can be pursued via follow-on Catalyst funding and/or bespoke integration engagements with merchants and Cardano-aligned venues that want a turnkey Hydra-backed PoS.

Future roadmap

- Continued **CNT coverage** (mint-policy whitelists, configurable per-merchant accepted-token sets).
- **Multi-merchant orchestration** patterns building on the per-head template.
- **Mobile-first PWA** polish on the customer page for live-venue use.
- Continued hardening from [docs/ops/incident-playbook.md](#) .

Permanent storage and forking instructions

- **Source code**: public GitHub, MIT licensed, fork-friendly.
- **Release artefacts**: attached to each tagged [GitHub Release](#) — downloadable + installable by the public.
- **Documentation**: GitHub Pages site, fully rebuildable from the cloned repo.
- **Forking instructions**: fork on GitHub, clone, follow the [README quick start](#) → [docs/ops/mainnet-deployment.md](#) . Required values (Blockfrost project key, `HP0S_KEY_SECRET` , etc.) are documented inline in the sanitised `.env.mainnet.example` . No proprietary services are required to build or run the test stack.

Project Completion Video (PCV)

The final community launch event was held at **Sova Bar, Da Nang**. Photos and videos from the launch are hosted on the project's shared media drive: [Community event videos and photos \(Google Drive\)](#). In-app product demo videos for the merchant + customer flows are committed in-repo under [docs/demo-videos/](#) .